



Jin Hyuk Kim

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EDUCATION

Hallym University

M.S. in Computer Engineering
AI-Powered Cheminformatics Lab (Advisor: Prof. Jonghwan Choi)
2025.03 – Present

Hallym University

B.S. in Big Data / Double Major, Smart IoT
2019.03 – 2025.02

RESEARCH INTERESTS

- **AI4Science**
 - Chemistry
 - Drug Discovery
- **Agentic AI for Molecular Design**
 - Synthesizable Molecular Generation
 - Multi-Objective Molecular Optimization
 - Planning and Routing
- **Molecular Representation Learning**
 - Multi-modal Fusion
 - Adaptive Fusion

HONORS & PATENTS

Outstanding Graduate Research Paper Award

Hallym University, 2026

Outstanding Presentation Paper Award

Korea Computer Congress (KCC), 2025

2 Patents Filed (2025): kinase selectivity prediction; LLM-based mutant binding-affinity prediction



Scan for full portfolio & publications

AI-Driven Molecular Discovery

I'm an M.S. student in Computer Engineering at Hallym University, advised by Prof. Jonghwan Choi in the AI-Powered Cheminformatics Lab. My interest is in chemistry, specifically how **Agentic AI**, **Generative AI**, and **Multi-modal Feature Fusion** can be applied to drug discovery. Across all of it, I care about building models that stay interpretable while they get more capable.

RESEARCH FOCUS AREAS

GENERATIVE AI

A Study on DDPM-based Molecular Generation and Semi-Supervised Learning for Improving the Performance of Optical Chemical Structure Recognition. *ASK, 2024 (Oral)*

OCSAug: Diffusion-based Optical Chemical Structure Data Augmentation for Improved Hand-drawn Chemical Structure Image Recognition. *J. Supercomputing, 2025*

Agentic Multi-objective Molecular Optimization via Dynamic Routing of Property-specific Editor Networks. *AI4Sci Korea, 2026*

AGENTIC AI

Agentic Multi-objective Molecular Optimization via Dynamic Routing of Property-specific Editor Networks. *AI4Sci Korea, 2026*

MULTI-MODAL QSAR

Pretrained Large Language Model-based Drug-Target Binding Affinity Prediction for Mutated Proteins (*co-author*). *J. KIISE, 2025*

A Study on Molecular Prompt Engineering Method for Large Language Model-based Molecular Property Prediction (*co-author*). *KCC 2025 (Outstanding Presentation Award)*

CUE: A Chemical Uncertainty-Aware Embedding Framework for Multi-modal Drug Selectivity Prediction. *Under Review, 2026*

SELECTED PUBLICATIONS

MORSE **AGENTIC AI** **GENERATIVE AI**

Agentic Multi-Objective Molecular Optimization · *AI4Sci Korea 2026, Poster*

A sequential-routing framework for multi-objective drug optimization, coordinating specialized “editor” models with an RL-trained router instead of relying on one-shot generation. This yields success rates of **0.73–0.84**, roughly **3–3.5×** higher than the prior state of the art, while staying interpretable through action-transition graphs.

CUE **MULTI-MODAL QSAR**

Multi-modal Drug Selectivity Prediction · *Under Review, 2026*

A multi-modal framework that predicts drug selectivity directly from molecular structure instead of stacking multiple binding-affinity predictions, avoiding compounding error. Molecular fingerprints and structure-image embeddings are fused through an uncertainty-aware weighting scheme, validated on a virtual-screening case study for EGFR double-mutant inhibitors.

OCSAug **GENERATIVE AI**

Diffusion-based Data Augmentation for Chemical OCSR · *J. Supercomputing, 2025*

The first diffusion-based augmentation method for hand-drawn chemical structure recognition, cutting image-distribution distance (FID) from 4+ down to **0.471** and improving recognition accuracy by up to **3.8×** across four OCR models.

PUBLICATIONS

1. **Kim, J.H.**, Choi, J. “OCSAug: Diffusion-based Optical Chemical Structure Data Augmentation for Improved Hand-drawn Chemical Structure Image Recognition.” *The Journal of Supercomputing*, 2025.
 2. **Kim, J.H.**, Choi, J. “Agentic Multi-objective Molecular Optimization via Dynamic Routing of Property-specific Editor Networks.” *AI4Sci Conference Korea*, 2026 (Accepted).
 3. **Kim, J.H.**, Kim, G.H., Park, H.J., Choi, J. “CUE: A Chemical Uncertainty-Aware Embedding Framework for Multi-modal Drug Selectivity Prediction.” 2026 (Under Review).
 4. Song, T.W., **Kim, J.H.**, Park, H.J., Choi, J. “Pretrained LLM-based Drug-Target Binding Affinity Prediction for Mutated Proteins.” *Journal of KIISE*, 2025.
- 3 additional publications (KCC 2025, ASK 2024, KIISE TCP 2026) are listed at jjjabcd.github.io/publications